

Saswat Meher

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Research Interests: Multimodal Representation Learning | Vision Language Models | Graph Machine Learning

EDUCATION

Indian Institute of Technology, Bombay

Mumbai, India

Master of Technology (M.Tech) in Computer Science and Engineering — CGPA: 9.69/10

Aug. 2022 – May 2024

National Institute of Technology, Rourkela

Rourkela, India

Bachelor of Technology (B.Tech) in Computer Science and Engineering — CGPA: 8.56/10

Aug. 2017 – May 2021

EXPERIENCE

Senior Software Developer

July 2024 – Present

Samsung R&D Institute India-Bangalore (SRI-B)

Bengaluru

- Developed a **foundational Vision-Language Model (VLM)** from the ground up to serve as the shared backbone for **5 production features** in the Samsung Gallery App.
- Executed end-to-end fine-tuning and on-device deployment of a **quantized TFLite VLM for image search**, driving an improvement of **30% in F1 Score** through optimized multimodal alignment.
- Architected and standardized a **VLM benchmarking framework**, unifying model evaluation across **5 teams** and significantly accelerating research-to-development cycles.
- Built an **automated annotation pipeline** using **ensemble of Multimodal LLMs**, accelerating the data labeling cycle from **months to days**.
- Trained a unified 16-language **X-EN translation model**, replacing legacy standalone versions to deliver a **10x boost** in resource efficiency with negligible loss in performance metrics.

Associate Software Engineer

Aug. 2021 – Aug. 2022

Veritas Technologies LLC

Pune, India

- Built a **process-management framework** for real-time monitoring and state-control of **long-running background processes** in Veritas File System.

PUBLICATIONS

Position: Graph Matching Systems Deserve Better Benchmarks.

ICML 2025

I. Roy, **Saswat Meher**, E. Jain, Soumen Chakrabarti, Abir De

Paper — Code

- Worked on experimental audit of graph similarity benchmarks, uncovering train-test leakage in standard datasets and implementing a novel data-augmentation framework to ensure robust model evaluation.

Graph Edit Distance with General Costs Using Neural Set Divergence

NeurIPS 2024

E. Jain, I. Roy, **Saswat Meher**, Soumen Chakrabarti, Abir De

Paper — Code

- A neural estimator for graph similarity that handles custom graph edit costs. Managed all benchmarking, including dataset curation and comprehensive accuracy and runtime analysis against every major baseline.

Predicting algal biochar yield using eXtreme Gradient Boosting (XGB)

Algal Research, 2020

S. Meher, A. Pathy, Balasubramanian P.

Paper — Code

- Executed an end-to-end ML pipeline using XGBoost to predict biochar yield, leveraging SHAP and Pearson correlation to identify temperature as the primary yield driver.

PROJECTS

Agentic Gallery

Code | Nov-2025

- Architected an **Agentic Gallery Assistant** using the **Koog framework** to orchestrate complex, multi-step user commands through **autonomous tool-calling**.
- Integrated a **Multimodal RAG pipeline** by using an **on-device VLM**, enabling privacy-centric, **natural language question-answering** across thousands of Gallery Images.

Multi Agent based Medical Assistant for Edge Devices

Report | Jan-2025

- Won **1st place** at the **Berkeley RDI LLM Agents Hackathon** for a multi-agent medical assistant.
- Developed a **multi-agent framework** for health monitoring of **4+ vitals and intelligent scheduling**.

TECHNICAL SKILLS

Languages: Python, C, C++

Libraries: PyTorch, LiteRT, torchao, vLLM, Ray, PyG, NetworkX